

Ashish Ranjan

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EXPERIENCE

C-DAC (Centre for Development of Advanced Computing), CINE – MeitY

IIT Guwahati Research Park, Assam
Onsite

May 2026 – Present
Software Engineering Intern

- Building **FORGE**, a multimodal deepfake threat intelligence system for the National Cyber Forensics Console, integrating text, image, and audio forensic pipelines with Explainable AI (SHAP)
- Engineered **deep learning models**: CNN-BiLSTM with 27-parameter LFCC feature extraction for synthetic speech detection; EfficientNetB0 + RF fusion with Grad-CAM for image forensics
- Applied **NLP & feature engineering**: stylometric analysis, TF-IDF, SBERT embeddings, and N-Gram scoring for AI-generated text detection with sentence-level XAI heatmaps

PROJECTS

F.O.R.G.E. – Forensic Observation and Recognition Gateway for Emerging Generative Exploits June 2026

- Built a **multimodal AI forensic system** analysing text, image, and audio evidence using explainable AI, forensic parameters, visual heatmaps, and automated legal report generation
- Implemented **deep learning pipelines**: EfficientNetB0 + Random Forest fusion for image forensics; CNN-BiLSTM with 27-parameter LFCC feature extraction for synthetic speech detection
- Applied **NLP and feature engineering**: stylometric analysis, TF-IDF, N-Gram diversity scoring, and SBERT semantic embeddings with SHAP-based XAI for sentence-level risk heatmaps
- Integrated **OCR-based document processing** (DOCX, PDF, TXT) and generated Grad-CAM visual heatmaps, waveforms, and spectrograms as forensic evidence artifacts

Digi-Darshan: Real-Time Crowd Management System | Python, Flask, WebSockets, PostgreSQL Sept. 2025

- Developed a real-time crowd monitoring system under **Smart India Hackathon (SIH)**, a national-level government initiative
- Implemented efficient **data pipelines and low-latency WebSocket updates** for high-throughput crowd data environments
- Designed scalable backend architecture ensuring system responsiveness and reliability under peak load conditions

ASHA Workers Healthcare Portal | Flask, MongoDB, PostgreSQL, REST APIs

Jan. 2026

- Building a **real-time healthcare platform** to improve medical service accessibility in remote regions of India
- Designed seamless **data exchange pipelines** between ASHA workers and Primary Health Centres for field-to-centre reporting
- Implementing **real-time monitoring dashboards** for service optimisation and performance tracking across health zones

EDUCATION

Sikkim Manipal Institute of Technology

B.Tech in Computer Science (Minor Specialisation: AI & ML)

Jul. 2024 – Aug. 2028
Rangpo, Sikkim

LEADERSHIP & ACTIVITIES

Smart India Hackathon (SIH)

2025

Team Lead

- Led a team to **qualify the internal round** of Smart India Hackathon, a national-level hackathon, driving end-to-end project execution, task distribution, and development workflow

Google Developer Student Clubs (GDSC)

2024

Volunteer

- Collaborated in community-driven technical initiatives associated with Google; contributed to peer learning and tech-based problem solving

SKILLS

Languages : Python, C++, JavaScript, SQL

AI / ML : Deep Learning, NLP, Feature Engineering, CNN-BiLSTM, EfficientNetB0, SBERT, TF-IDF, SHAP (XAI), LFCC, Predictive Modelling, Neural Networks, Generative AI, LLMs

Digital Forensics : Audio Forensics, Image Forensics, Text Forensics, OCR, Grad-CAM, Spectrogram Analysis

Backend & Web : FastAPI, Flask, REST APIs, Node.js, PostgreSQL, MongoDB

Tools & Systems : Linux, Git, Google Colab, Kaggle, GNU Make, VS Code, GitHub

CERTIFICATIONS

Crash Course on Python – Coursera/Google (2025) GenAI: Unlock Foundational Concepts – Google Cloud (2026)

Advanced React – Coursera/Meta (2025) Advanced Java Programming – Coursera/Meta (2025)